Few-Shot Aerial Image Semantic Segmentation Leveraging Pyramid Correlation Fusion

Wei Ao, Shunyi Zheng, Yan Meng, and Zhi Gao

***Abstract*— Few-shot semantic segmentation (FSS) has gained significant attention due to its ability to segment novel objects using only a limited number of labeled samples, thereby addressing the problem of overfitting caused by a lack of training data. Although this technique is widely studied in the field of computer vision, there are few methods for remote-sensing images. Prevalent FSS methods can achieve remarkable results for natural images, but they are difficult to apply to remote-sensing image processing because existing methods rarely take into consideration the large-scale and resolution differences in remote-sensing images. Consequently, it is hard for them to obtain correct semantic guidance from a few annotated remote-sensing images. To tackle these problems, this article proposes the pyramid correlation fusion network (PCFNet) to promote the ability to mine helpful information by calculating multiscale pixel-wise semantic correspondence. Particularly, the dual-distance correlation (DDC) module is designed to simultaneously compute the cosine similarity and Euclidean distance between query features and support features, producing adequate guidance information to determine the category of each pixel. Moreover, to improve segmentation accuracy for small objects, the scale-aware cross-entropy loss (SACELoss) is introduced to dynamically assign loss weights according to the actual sizes of objects. This enables smaller objects to be assigned larger weight values and thus receive more attention during training. Comprehensive experiments on both the iSAID-5^i and DLRSD-5^i datasets demonstrate that our method outperforms state-of-the-art FSS methods. Our code is available at https://github.com/TinyAway/PCFNet.**

***Index Terms*— Distance correlation, few-shot semantic segmentation (FSS), meta-learning, remote-sensing image processing, semantic correspondence.**

I. Introduction

SEMANTIC segmentation is a fundamental task in remote-sensing image processing and has been widely used in building extraction [1], [2], [3], change detection [4], [5], [6], and land cover mapping [7], [8], [9]. Benefiting from the rapid development of deep learning [10], remote-sensing image semantic segmentation has made great progress in recent years. Currently, mainstream semantic segmentation methods tend to adopt an encoder–decoder architecture to execute end-to-end training mode. These methods perform well when sufficient training samples are available, as proven by numerous studies [11], [12], [13], [14]. However, their good performance relies on extensive and well-annotated datasets such as iSAID [15] and Gaofen image dataset (GID) [16]. Unfortunately, these large-scale datasets require significant human effort to annotate images and are not available in many practical applications. When the training data is limited, the performance of these approaches deteriorates severely.

Few-shot semantic segmentation (FSS) has become increasingly popular in recent years as a way to address the problem of accuracy decline caused by a lack of training samples [17]. Unlike traditional semantic segmentation algorithms, FSS is designed to segment unseen objects using meta-learning [18]. FSS models also can be trained end-to-end, with each input consisting of a set of images referred to as an episode, which includes a query image to be segmented and one or few support images with annotations. Importantly, the query image and the support images contain the same category to be identified. During training, FSS models can extract useful clues from the support images to segment the query image. Therefore, the main challenge of FSS is to accurately mine the latent correlation between the query image and the support images. Current FSS methods typically use dense comparisons between query and support features to achieve this [19], [20], [21], which have shown impressive performance in natural images. However, remote-sensing images differ significantly from natural images in several ways. For example, the scale of objects in urban scenes varies greatly due to the numerous sources of image acquisition [22], [23]. On the one hand, there are numerous small objects such as vehicles and ships. On the other hand, targets like sports fields are so large that can only be presented partially in a 256×256 image. Furthermore, remote-sensing image datasets often suffer from extremely uneven categories. Especially, the number of buildings is far more than that of bridges. As a result, directly applying existing FSS methods to remote-sensing images cannot produce satisfactory results.

Recently, innovative approaches leveraging multitask learning [24], [25] and meta-learning [26], [27], [28] have emerged to address the challenge of few-shot hyperspectral image (HSI) classification. These methods exhibit the capability to identify unfamiliar objects within HSI datasets, even when provided with only limited labeled samples. Nevertheless, their performance can be hindered when compared to red, green, and blue (RGB) aerial images due to the latter's higher spatial resolution and greater variety of ground objects. Consequently, even cutting-edge few-shot learning (FSL) techniques struggle

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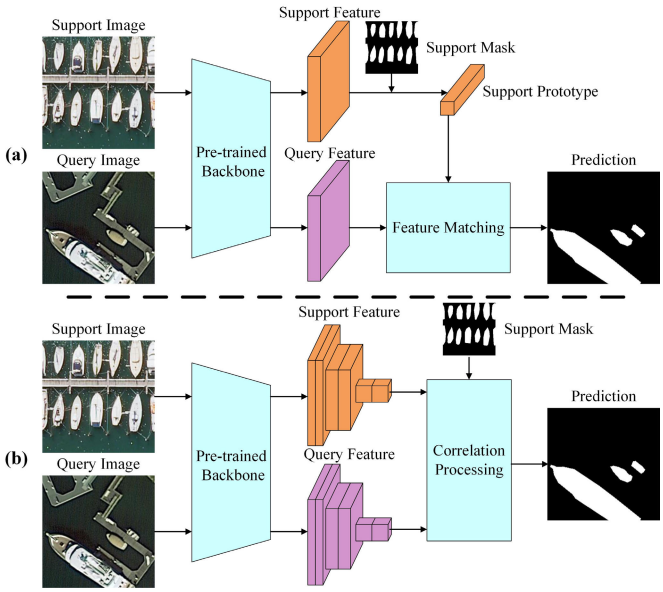


Fig. 1. Comparison between the FSS methods based on (a) prototype matching and (b) pixel-wise correspondence. Prototypical methods calculate a support prototype to make dense comparisons with query features. Methods based on pixel-wise correspondence attempt to compute the correlation of pixels between the query and support features.

to extract robust features from such imagery. To address this challenge, a metric-learning-based approach has been introduced to facilitate accurate class correspondence between query and support images. For example, Yao et al. [29] proposed a detailed matching module to establish precise pixel-level correspondence and introduced a scale-aware focal loss (SAFLoss) to improve recognition of small objects. This method can achieve more promising performance than state-of-the-art FSS approaches. Similarly, Wang et al. [30] proposed a feature pyramid comparison network that can acquire scale-balanced prototypes from multilevel support features. This approach mitigates the issue of unclear understanding of the scale variations of geographic objects. Both of these methods adopt a prototype-based paradigm [as depicted in Fig. 1(a)], relying on accurate category prototypes extracted from support features. However, a notable drawback lies in their inherent loss of fine-grained details, potentially resulting in subpar performance when query and support images significantly differ. This limitation stems from their reliance on prototype matching, which may yield unreliable guidance information for subsequent segmentation processes. Therefore, to overcome this limitation, we attempt to generate more guidance information by computing multilevel pixel-wise semantic correlation. The model processes the corresponding level feature maps of the query image and the support image, as shown in Fig. 1(b), to produce multiscale correlation maps. Then, the category of each pixel is determined by fusing the correlation maps of all levels.

In this article, we propose the pyramid correlation fusion network (PCFNet) to enhance the FSS of remote-sensing images. We first extract multiscale features of the query image and support images using a backbone network. Then a dual-distance correlation (DDC) module is designed to compute dual-correlation maps (DCMs) between the query features and support features leveraging both cosine similarity and Euclidean distance. Through a pyramid correlation decoder, the correlation maps calculated from feature maps of various scales are fused to produce precise prediction guidelines. Moreover, inspired by SAFLoss [29], we propose a more robust scale-aware cross-entropy loss (SACELoss) to strengthen the recognition of small objects. It can automatically assign higher loss weight to small objects and lower loss weight to large objects. Consequently, the model will pay more attention to tiny hard-parsed objects during training. Although remote-sensing semantic segmentation datasets such as the Potsdam and Vaihingen datasets [31] are widely used, they may not be sufficient for FSS models that require extensive training on base classes and evaluation of new classes. To address this issue, we have developed a new FSS dataset called DLRSD-5^i, which is derived from the dense labeling remote sensing dataset (DLRSD) [59] and consists of 15 categories. Our dataset provides a more diverse and comprehensive set of images for training and evaluation, which can help improve the performance of few-shot aerial image semantic segmentation models. To demonstrate the effectiveness of our method, we conducted comprehensive experiments on the iSAID-5^i [29] and DLRSD-5^i datasets under 1-shot and 5-shot settings. Our model outperforms all state-of-the-art FSS methods. The main contributions of our paper are summarized as follows.

1) We propose a DDC module to greatly explore the semantic correspondence between the query features and the support features.
2) We develop an elaborate SACELoss that improves model performance by accommodating objects of various scales during training.
3) We construct a new benchmark for few-shot remote-sensing image semantic segmentation, called DLRSD-5^i, to thoroughly assess the performance of our model.
4) We conduct extensive experiments on the iSAID-5^i and DLRSD-5^i datasets and show that the proposed PCFNet outperforms other FSS methods and achieves state-of-the-art results under both 1-shot and 5-shot settings.

II. Related Work

A. Remote-Sensing Image Semantic Segmentation

Semantic segmentation is a fundamental task in image comprehension that requires dense prediction of every pixel. Recently, great progress has been made in semantic segmentation due to the emergence of fully convolutional networks [32]. In the field of remote sensing, semantic segmentation is widely used to identify geographic objects such as buildings and vegetation. In general, popular semantic segmentation techniques such as U-Net [33] and PSPNet [34] can achieve promising results in remote-sensing images. However, for specific application scenarios such as building and road extraction, practical techniques can be applied to improve prediction accuracy. For instance, Xu et al. [35] utilized the normalized differential vegetation index (NDVI) and the normalized digital surface model (NDSM) as additional inputs to enhance the recognition of buildings. Batra et al. [36] designed an orientation learning

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task that guides the model to extract consecutive and topologically accurate road networks. Guo et al. [37] introduced a boundary refinement module that improves the accuracy of building footprints by determining the direction of each pixel toward the center of the object to which it belongs. However, regardless of their effectiveness, these methods rely heavily on high-quality labeled datasets. In contrast, our model is built on meta-learning and is capable of recognizing new objects with only a few annotated images.

B. Few-Shot Learning

FSL is a learning paradigm that seeks to develop models capable of learning a novel task with a small amount of labeled data. Current research studies are primarily based on meta-learning, which involves training models on several learning tasks to gain the ability to solve new ones. These works can be further divided into two categories: gradient- and metric-learning-based methods. Gradient-based approaches aim to find the optimal gradient descent direction to achieve the best performance quickly. For example, Andrychowicz1 et al. [38] regarded the optimization algorithm as a learnable task, thus the optimizer can be obtained through training. Model-agnostic meta-learning (MAML) [39] attempts to optimize the trained initial parameters one or more times to quickly adapt to a new task. Metric-learning-based methods, on the other hand, identify novel objects by computing the feature correlations between the query image and the support images. For instance, the relation network [40] and the prototypical network [41] attempt to learn nonlinear mappings that represent the features of each class through embedding, so that the query image can be easily classified by comparing these embeddings with query features. It is worth noting that metric-learning-based methods have gained widespread attention in the task of few-shot scene classification of remote-sensing images [42], [43], [44]. These methods aim to discover a reliable distance metric, such as the Euclidean distance [45] and scaled cosine similarity [46], to uncover the latent relationship between the query and support images. Although both of these methods have demonstrated promising results in few-shot classification tasks, FSS is a more challenging problem. Applying FSL methods directly to address this task does not yield satisfactory results, as previous studies have shown [19], [21].

C. Few-Shot Semantic Segmentation

FSS can be regarded as a pixel-wise FSL task, which has attracted considerable attention in recent years. To guide the segmentation of the query image, several methods have been proposed to find precise prototypes for the support features, inspired by the success of prototypical networks [41]. For instance, PANet [19] utilizes prototype alignment regularization to obtain comprehensive prototypes, whereas PFENet [47] computes a prior knowledge mask from high-level feature maps to provide reliable guidance for parsing. PMMs [48] generate multiple prototypes for different regions using an expectation–maximization algorithm to address the semantic ambiguity issue. However, these methods struggle when the query image and support images vary significantly in scale or shape. Therefore, recent research studies have explored the potential of multiscale semantic correspondence. For example, hypercorrelation squeeze network (HSNet) [49] proposed to utilize an efficient 4-D convolution to process pyramidal correlation maps generated from feature maps at different scales. VAT [50] designed a 4-D convolutional Swin Transformer to solve problems caused by limited local context near patch boundaries and low inductive bias. FECANet [51] utilized a feature enhancement module to filter out noise in the correlation map that arises from local similarity and intraclass diversity. However, these methods directly process the original 4-D pixel-wise correlation tensor, which can be inefficient due to its sparsity. To address this issue, this article proposes using a 2-D DCM instead of the 4-D pixel-wise correlation tensor to reduce redundant computations.

III. Methodology

The architecture of the proposed PCFNet is illustrated in Fig. 2. Our approach begins with 1-shot segmentation in Sections III-B–III-E, and we then explain how we extend it to few-shot segmentation in Section III-F. First, a pretrained backbone is used to extract multilevel features from a pair of query and support images. These features and the support mask are then fed into a DDC module to compute the semantic relevance of the corresponding hierarchical features. As the DDC module calculates both cosine similarity and Euclidean distance, each feature map pair generates two distance correlation maps (cosine correlation map and Euclidean correlation map). Next, a pyramid correlation decoder is utilized to fuse multiple distance correlation maps and extract useful information for producing the segmentation mask. Finally, we design a SACELoss to improve the recognition ability of small objects by assigning lower loss weight to large targets and higher loss weight to small targets, thereby enabling the model to focus on the identification of difficult-to-segment small objects.

A. Task Descriptions

FSS is a task that aims to train a model on base categories and segment novel categories in query images by learning from a few annotated support images. To accomplish this, we adopt the widely used meta-learning method called episodic training [52]. In this method, all categories are divided into two nonoverlapping class sets, namely C_{train} and C_{test}, and the model is trained on classes in C_{train} and evaluated on classes in C_{test}. In the N-shot segmentation setting, each learning episode consists of a query set $Q = \{I^q, M^q\}$ and a support set $S = \{(I_i^q, M_i^q)\}_{i=1}^N$ that contain N support images, where I and M represent an image and its corresponding ground truth, respectively. During training, for each epoch, a category in C_{train} is randomly sampled, and $N+1$ images containing this category are selected from the training set D_{train} to form an episode. The goal of the learning task is to find a model that can accurately map the query–support pair I^q, S to the query mask M^q. Finally, the trained model is tested on images in the test set D_{test} that contain the categories in C_{test}.

B. Multilevel Feature Extraction

In this section, we utilize a pretrained backbone network, such as ResNet50 [53] pretrained on ImageNet [54], to extract

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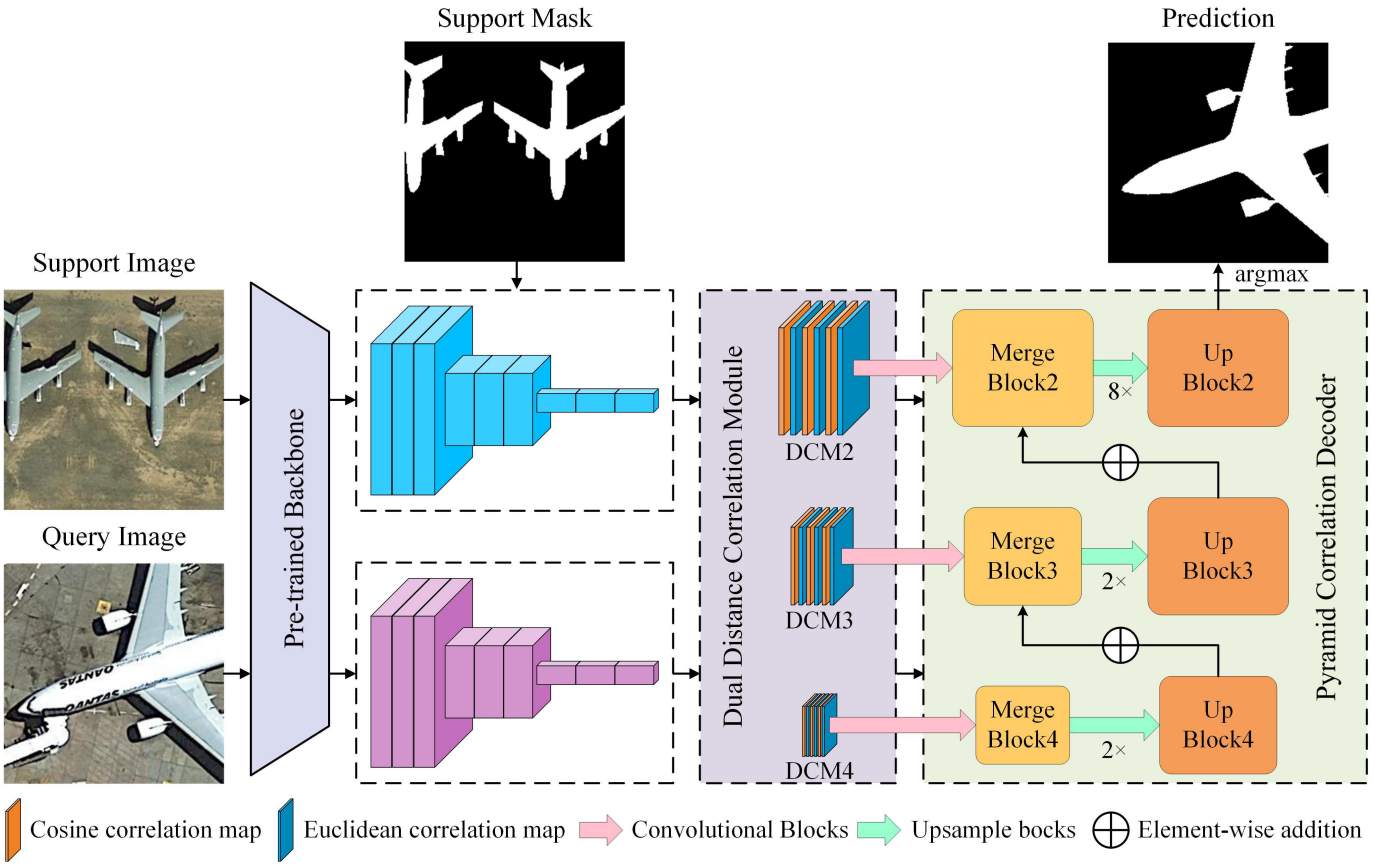


Fig. 2. Overall architecture of PCFNet. The model consists of three parts: a pretrained backbone network, a DDC module, and a pyramid correlation decoder. The pretrained backbone network is responsible for extracting multiscale features from both the query and support images. Subsequently, the DDC module generates DCMs from these features. Finally, the pyramid correlation decoder combines the DCMs from three different scales to produce the prediction mask.

multilevel features from a pair of images consisting of a query image $I^q \in \mathbb{R}^{3\times H\times W}$ and a support image $I^s \in \mathbb{R}^{3\times H\times W}$. Following HSNet [49], we collect all the feature maps produced by the last three convolutional blocks (blocks 2–4). This process is expressed as follows:

$$F^q = \left\{ \left(F^q_{b,l} \right)_{l=1}^{L} \right\}_{b=2}^{B} \tag{1}$$

$$F^s = \left\{ \left(F^s_{b,l} \right)_{l=1}^{L} \right\}_{b=2}^{B} \tag{2}$$

where B and L denote the block number and the bottleneck number of the backbone, respectively. For instance, $F^q_{3,2}$ represents the query feature maps generated by the second bottleneck of block 3. Then, they are fed into the DDC module along with the support mask $M^s \in \{0,1\}^{H\times W}$ to produce DCMs.

C. DDC Module

As illustrated in Fig. 3, the DDC module aims to produce two correlation maps from each pair of query features and support features by computing their cosine similarity and Euclidian distance, simultaneously.

Previous studies have demonstrated the effectiveness of calculating cosine similarity for FSS [19], [49]. For each support feature map $F^s_{b,l} \in \mathbb{R}^{C_b\times H_b\times W_b}$, it is first masked with the corresponding bilinearly interpolated support mask $M^s_b \in 0, 1^{H_b\times W_b}$ as follows:

$$F^{\text{ms}}_{b,l} = F^s_{b,l} \odot M^s_b \tag{3}$$

where C_b, H_b, and W_b refer to the channel dimension, height, and width of the feature maps produced by block b, respectively. $\odot$ represents the Hadamard product. Then $F^{\text{ms}}_{b,l}$ is considered as foreground features and can be used together with the query features to calculate cosine similarity

$$\text{CS}_{b,l} = \frac{F^{q\ T}_{b,l} \cdot F^{\text{ms}}_{b,l}}{\left\| F^q_{b,l} \right\| \cdot \left\| F^{\text{ms}}_{b,l} \right\| + \varepsilon} \tag{4}$$

where ε is a small constant. As can be observed from the above equation, when the target in the supported image is relatively small, most of the values in $\text{CS}_{b,l} \in [0,1]^{H_bW_b\times H_bW_b}$ are 0. Directly calculating the mean value like MSANet [55] will result in very small values and thus cause them to be misjudged as backgrounds. To address this problem, we refer to the operation of masked average pooling (MAP) [19] where we only average the nonzero pixels, instead of averaging over all pixels

$$C^{\cos}_{b,l} = \frac{\sum_{i=1}^{H_bW_b} CS_{b,l,i}}{\sum_{i=1}^{H_bW_b} M^s_{b,i}}. \tag{5}$$

Therefore, $C^c_{b,l} \in [0,1]^{H_b\times W_b}$ represents the cosine correlation map we compute.

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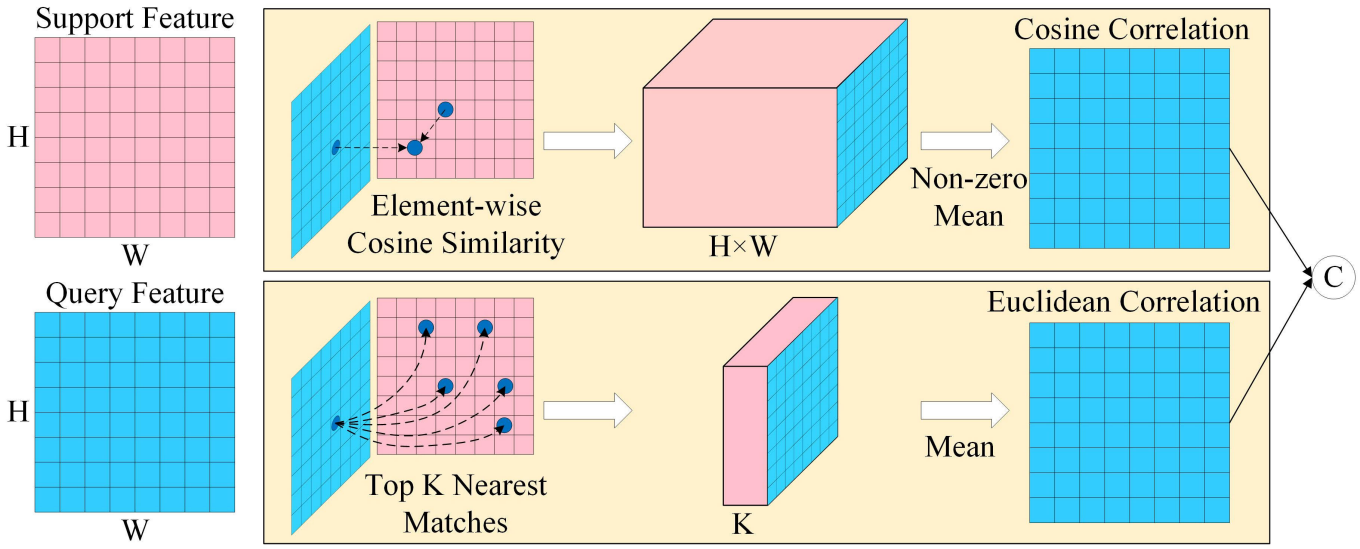


Fig. 3. Illustration of the DDC module. In this module, we perform the simultaneous computation of cosine similarity and Euclidean distance between the query features and the support features to produce two correlation maps (cosine correlation map and Euclidean correlation map). To produce the cosine correlation map, we calculate the element-wise cosine similarity between the query and support features. Next, we average the values corresponding to the foreground area. To create the Euclidean correlation map, we average the ground truth corresponding to the K supporting features that are closest to each pixel in the query feature map. Finally, we concatenate the two correlation maps along the channel dimension to produce the DCM.

Although the use of the cosine correlation can achieve good performance, it is limited by redundant information and easily confused features. Therefore, we propose using only the top K matched features to construct the Euclidean correlation. For each pixel of a query feature $F^q_{b,l}$, we identify the locations of its K nearest neighbors $\text{KNN} = \{(x_i, y_i)\}_{i=1}^{K}$ in $F^s_{b,l}$. Then, we generate its Euclidean correlation map by averaging the values of these K positions in M^s_b

$$C^E_{b,l} = \frac{\sum_{i=1}^{K} M^s_{b,l,\text{KNN}_i}}{K}. \tag{6}$$

Finally, we concatenate all cosine correlation maps and Euclidian correlation maps for each block

$$C_b = \left\{ C^{\cos}_{b,l} \copyright C^E_{b,l} \right\}_{l=1}^{L} \tag{7}$$

where © represents concatenation along channel dimension. Therefore, $C_b \in [0, 1]^{2L_b \times H_b \times W_b}$ represents the final DCM of block b.

D. Pyramid Correlation Decoder

In this module, we will be fusing and decoding three DCMs in a structure that is similar to FPN [56]. First, each correlation map undergoes conversion to the same channel dimension using a decoding block. The decoding block is composed of a series of 2-D convolution, group normalization, and rectified linear unit (ReLU) functions. Next, the two outputs from adjacent pyramid layers are merged through element-wise addition after double upsampling of the deeper layer output. Furthermore, we obtain a two-channel map $\hat{P} \in \mathbb{R}^{2\times H\times W}$ via upsampling the merged output once by a factor of 8. Ultimately, the prediction mask $P \in \{0, 1\}^{H\times W}$ is obtained by applying an argmax function on the channel dimension.

E. Scale-Aware Cross-Entropy Loss

In remote-sensing image semantic segmentation, extracting tiny objects with precision is quite a challenging task. To address this issue, Yao et al. [29] proposed a SAFLoss that concentrates on the accurate parsing of small objects, resulting in significant improvements. This approach determines the object size by calculating the proportion of all foreground pixels in the image. The smaller an object, the higher its loss value. However, this method only applies to single objects and fails to handle scenarios with more than one tiny object in an image, such as numerous cars in a parking lot. To overcome this limitation, we designed a SACELoss inspired by the weighted cross-entropy loss (CELoss). This new loss function calculates the proportion of the area of each individual foreground object and assigns a weight value to all its pixels based on its proportion. Specifically, for each pixel of the query mask M^q, its weight is computed as follows:

$$W_i = \begin{cases} 1 - \log\left(\dfrac{A_i}{HW}\right), & \text{if } M^q_i = 1 \\ 1, & \text{if } M^q_i = 0 \end{cases} \tag{8}$$

where A_i denotes the area of the object to which pixel i belongs. Using (8), the smaller the object is, the larger the weight value it gets. To summarize, the SACELoss is defined as follows:

$$L = \frac{1}{HW} \sum_{i=1}^{HW} \left\{ -W_i \left[M^q_i \cdot \log P_i + \left(1 - M^q_i\right) \cdot \log(1 - P_i) \right] \right\}. \tag{9}$$

Note that our loss function calculates weights based on the size of individual objects, rather than treating all small objects as a single large target as the SAFLoss does. This approach allows our loss function to focus on both the size and details

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TABLE I
TEST CLASSES FOR EACH FOLD OF DLRSD-5^i

Dataset	Test classes
Fold0	airplane, bare soil, buildings, cars, chaparral
Fold1	court, dock, grass, mobile home, pavement
Fold2	sand, ship, tanks, trees, water

of individual objects, even when there are many small objects in the image. We discuss the superiority of our loss function and its contrast with the SAFLoss in Section IV-D.

F. Extending to Few-Shot Setting

We have presented the model under a 1-shot scenario, and there are two methods to extend the model to the N-shot setting. The first method employs a training-free voting strategy, which follows the approach used in HSNet [49]. This method divides the N-shot segmentation into N 1-shot segmentations, enabling the final prediction to be produced by voting on each pixel using N segmentation masks. This is a flexible approach that can be used for inference without requiring additional model training. The second method involves taking the average of N correlation maps at each block, which serve as the new correlation maps before being fed into the pyramid correlation decoder. This approach requires training a separate model for the N-shot segmentation task. We utilized both methods and made a comparison of their performances, which are presented in detail in Section IV-D.

IV. EXPERIMENTS

A. Datasets and Evaluation Metrics

In our experiments, the iSAID-5^i [29] and DLRSD-5^i datasets were utilized as the benchmark to evaluate the model. The iSAID-5^i dataset is built from the iSAID dataset [15] and consists of 15 classes that are evenly divided into threefold. All images have a size of 256 × 256 pixels and are cropped from larger images. Similarly, the DLRSD-5^i dataset is constructed based on DLRSD [59], which contains 17 classes. Likewise, we select 15 classes and evenly split them into threefold, and the specific details of class splits are presented in Table I. Since the original dataset did not have a predefined split between the training and test sets, we randomly divided the data in a 7:3 ratio to create these two subsets. Following FWB [60], we perform cross-validation over three folds; for each fold, its five classes are designated for testing, and the remaining ten classes from other folds are used for training.

To quantitatively evaluate the model's performance, we use mean intersection over union (mIoU) as our primary evaluation metric. This metric is calculated by averaging the intersection over union (IoU) values of all categories. Meanwhile, foreground–background IoU (FB-IoU) is also calculated to measure the ability of the model to extract class-agnostic information

$$\text{mIoU} = \frac{1}{C}\sum_{c=1}^{C}\text{IoU}_c \tag{10}$$

$$\text{FB}-\text{IoU} = \frac{1}{2}\left(\text{IoU}_f + \text{IoU}_b\right) \tag{11}$$

where C is the number of classes and IoU_c represents the IoU value of class c. IoU_f and IoU_b denote the foreground and background IoU values in the target fold, respectively. To ensure a stable assessment, we randomly sample 5000 and 1000 episodes for iSAID-5^i and DLRSD-5^i, respectively, during testing on each fold. We then take the average of the results under five different random seeds as the final mIoU/FB-IoU.

B. Implementation Details

In our experiments, VGG16 [61] and ResNet50 [53] that were pretrained on ImageNet [54] were adopted as the backbone networks. All feature maps produced by the final three convolutional blocks of the backbone are collected. To prevent these networks from learning class-specific knowledge of the training data, we froze their parameters during the training process. The parameter K used in the DDC module was set to 10 and is further studied in the ablation study. Our networks were implemented with PyTorch and optimized using Adam with a learning rate of 1e-4. Except for random vertical and horizontal flips, no additional data augmentation method was used. We employed OpenCV to extract all objects from the ground truth, subsequently determining their weight values by leveraging the actual area of each individual object. For each fold, we used a batch size of 4 and trained the model for 100 000 and 50 000 iterations for iSAID-5^i and DLRSD-5^i, respectively. Due to the highly imbalanced categories in both datasets (e.g., the number of cars is much larger than bridges), we balanced the categories of samples during training to prevent the model from learning class-specific knowledge. Rather than using every image in the training set as the query image, we randomly selected a category and then sampled an image from the training set containing that category as the query image. To testify to the superiority of the proposed PCFNet, both 1-shot and 5-shot segmentation were conducted in the three folds of iSAID-5^i and DLRSD-5^i, and they were compared with other state-of-the-art FSS methods.

C. Results and Analysis

Several state-of-the-art FSS methods such as PFENet [47], MLC [57], HSNet [49], DCAMA [58], and SDM [29] were selected to compare with our model under the same experimental settings. Table II presents the detailed performance of these methods with two backbone networks in the experiments on the iSAID-5^i dataset. It is obvious that our model significantly outperforms all its competitors by a large margin in both 1-shot and 5-shot experiments. Specifically, when comparing the mean mIoU values with VGG16, our method obtains 9.92% and 15.45% mIoU improvements over DCAMA in the 1-shot and 5-shot segmentation, respectively. Simultaneously, using ResNet50 as the backbone, our method obtains 10.15% and 16.07% mIoU improvements over HSNet in the 1-shot and 5-shot segmentation, respectively. Additionally, we obtained the same results in terms of mean FB-IoU. Compared with DCAMA, our proposed PCFNet with VGG16 and ResNet50

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TABLE II

MIOU AND FB-IOU RESULTS ON THREE FOLDS OF iSAID-5^i COMPARED WITH STATE-OF-THE-ART FSS METHODS. † REPRESENTS THE RESULTS REPORTED BY THE ORIGINAL PAPER. THE BEST PERFORMANCES ARE HIGHLIGHTED IN BOLD

Methods	1-shot					5-shot				
	Fold0	Fold1	Fold2	Mean	FB-IoU	Fold0	Fold1	Fold2	Mean	FB-IoU
VGG16										
PFENet [47]	28.57	23.38	27.77	26.57	50.59	28.02	27.23	28.14	27.80	51.66
MLC [57]	25.76	20.59	26.60	24.32	49.52	37.21	26.72	35.87	33.27	55.26
HSNet [49]	23.41	21.61	22.01	22.34	50.21	28.76	27.15	25.90	27.27	54.81
DCAMA [61]	28.47	15.79	26.87	23.71	45.83	31.60	20.08	31.33	27.67	50.44
SDM† [29]	29.24	20.28	**34.73**	28.26	-	36.33	27.98	42.39	35.77	-
PCFNet (Ours)	**39.93**	**27.42**	33.55	**33.63**	**55.89**	**50.73**	**34.38**	**44.25**	**43.12**	**61.72**
ResNet50										
PFENet [47]	28.57	23.38	27.77	26.57	50.59	28.02	27.23	28.14	27.80	51.66
MLC [57]	28.53	21.33	32.44	27.43	50.67	38.44	28.58	37.55	34.86	55.82
HSNet [49]	30.82	22.91	32.94	28.89	56.13	32.94	29.73	40.76	34.48	59.56
DCAMA [61]	32.73	18.25	28.74	26.57	51.92	35.16	20.25	35.87	30.43	49.83
SDM† [29]	34.29	22.25	32.62	30.72	-	39.88	30.59	45.70	38.72	-
PCFNet (Ours)	**42.94**	**27.17**	**40.06**	**36.72**	**57.78**	**52.97**	**34.91**	**51.62**	**46.50**	**63.59**

TABLE III

MIOU AND FB-IOU RESULTS ON THREE FOLDS OF DLRSD-5^i COMPARED WITH STATE-OF-THE-ART FSS METHODS. THE BEST PERFORMANCES ARE HIGHLIGHTED IN BOLD

Methods	1-shot					5-shot				
	Fold0	Fold1	Fold2	Mean	FB-IoU	Fold0	Fold1	Fold2	Mean	FB-IoU
VGG16										
PFENet [47]	17.08	15.85	13.70	15.54	38.68	21.25	17.40	18.10	18.92	39.97
MLC [57]	25.93	35.31	35.63	32.29	39.84	30.79	39.04	40.97	36.93	41.74
DCAMA [61]	20.48	7.42	10.53	12.81	38.64	25.25	10.74	13.03	16.34	40.57
HSNet [49]	20.26	25.80	16.95	21.00	43.56	26.02	29.92	27.07	27.67	47.40
PCFNet (Ours)	**30.75**	**36.38**	**40.76**	**35.96**	**45.72**	**35.64**	**41.57**	**51.18**	**42.80**	**51.53**
ResNet50										
PFENet [47]	21.98	20.67	11.17	17.94	38.82	24.12	24.76	16.01	21.63	40.39
MLC [57]	26.38	36.05	34.65	32.36	40.06	31.22	40.33	43.03	38.19	45.65
DCAMA [61]	12.68	13.01	10.60	12.10	36.43	14.37	15.67	15.24	15.09	39.32
HSNet [49]	24.61	29.45	28.24	27.43	45.45	29.67	31.95	36.93	32.85	48.29
PCFNet (Ours)	**31.09**	**37.52**	**41.89**	**36.83**	**46.65**	**36.67**	**42.70**	**54.20**	**44.52**	**52.40**

shows performance improvements of 10.06% and 5.68% in the 1-shot setting, respectively, and 11.28% and 13.76% in the 5-shot setting, respectively.

In terms of experiments conducted on DLRSD-5^i, our proposed PCFNet has demonstrated an exceptional ability to maintain the highest segmentation accuracy, as evidenced by Table III. Specifically, our method outperforms all competitors in both 1-shot and 5-shot experiments, regardless of whether VGG16 or ResNet50 is used as the backbone network. It is noteworthy that traditional prototype-based methodologies, such as PFENet, struggle to perform effectively on this dataset, primarily due to their inability to harness the wealth of texture information inherent in the DLRSD-5^i dataset. Furthermore, despite its status as an advanced FSS method, DCAMA exhibits surprisingly underperformance on this dataset. This phenomenon largely owing to the inadequacy of training data when compared to the iSAID-5^i dataset. These results suggest that by integrating the elaborate DDC module and SACELoss, the overall performance of the model can be significantly enhanced.

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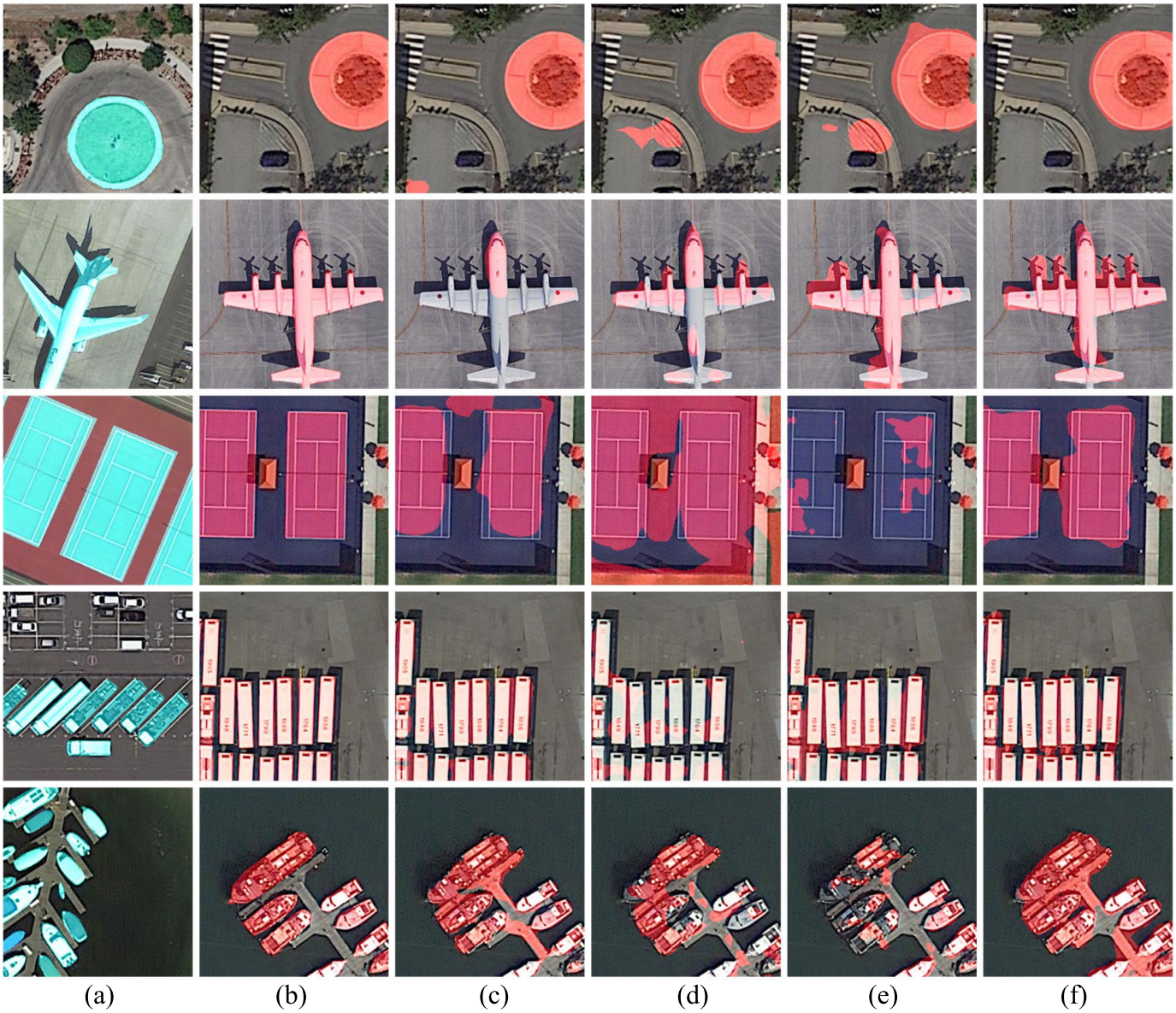


Fig. 4. Qualitative results of our model compared with state-of-the-art FSS methods. (Left–right) (a) Support image and its ground truth, (b) query image and its ground truth, segmentation results of (c) PFENet, (d) MLC, (e) HSNet, and (f) our PCFNet.

To clarify the advantages of our model compared to other models, we have selected several support–query pairs from the iSAID-5^i dataset and visualized the segmentation results of our model and its competitors. As shown in Fig. 4, our model can achieve accurate segmentation results for objects of various sizes. Moreover, our model employs a multiscale feature extraction process to derive semantic correspondences for each pixel, resulting in more complete segmentation results compared to other methods. It is worth noting that although HSNet also utilizes semantic correspondences, it solely relies on cosine similarity and is prone to being misled by background noise. However, our method uses Euclidean distance and top-K nearest matches to effectively filter out background interference, thereby improving the overall segmentation performance.

Finally, we conducted a comparison of the computational efficiency of different models by evaluating their number of learnable parameters, per-episode inference time, and multiply-accumulate operations (MACs). Specifically, we assessed the 1-shot performance of four models using ResNet50 as the backbone, with all evaluations being carried out on an NVIDIA GeForce GTX TITAN X. Table IV provides a summary of the results. Notably, our model has only 1.4 M trainable parameters, which is significantly lower than other methods. Additionally, our model requires only 2.0G MACs, making it the most computationally efficient among the models evaluated. Although the execution speed of the KNN algorithm in the DDC module limits the inference speed of our method slightly compared to HSNet (42.4 versus 28.8 ms), it is still faster than PFENet and DCAMA. Overall, our model stands out not only for its superior accuracy, but also for its impressive computational efficiency.

To summarize, the superiority of the proposed PCFNet can be analyzed from three key aspects. First, conventional prototype-based methods struggle to obtain accurate target prototypes due to large-scale differences and small interclass

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TABLE IV
COMPARISON BETWEEN FIVE DIFFERENT MODELS IN MODEL SIZE, PER-EPISODE INFERENCE TIME, AND MACS

Method	Learnable params (M)	Inference times (ms)	MACs(G)
PFENet	10.8	49.1	62.3
MLC	27.7	**23.1**	8.7
HSNet	2.6	28.8	3.1
DCAMA	52.1	44.2	17.7
PCFNet (Ours)	**1.4**	42.4	**2.0**

TABLE V
IMPACT OF THE TWO COMPONENTS IN THE DDC MODULE

Cosine	Euclidian	Fold0	Fold1	Fold2	Mean	FB-IoU
✓		38.98	24.92	39.39	34.43	56.94
	✓	40.57	26.82	37.61	35.00	56.76
✓	✓	**42.94**	**27.17**	**40.06**	**36.72**	**57.78**

variance in remote-sensing images, while our DDC module calculates multiscale pixel-wise distance correlation to provide sufficient guidance information for target identification in query images. Second, small objects are often challenging to extract in remote-sensing image interpretation, but our elaborated SACELoss can dynamically assign weights according to the actual size of targets during training, helping the model focus on small targets. Finally, it is commonly known that large parameter quantity can easily result in the loss of generalization power of the model [62]. Our model uses only cosine similarity and Euclidean distance calculation without adding additional parameters, which protects the model from overfitting during training.

D. Ablation Study

To evaluate the impact of each component in our method, we conducted a series of ablation tests on the iSAID-5^i dataset. The DDC module, top-K nearest matches, loss functions, and few-shot strategies were analyzed. Following are the detailed test results.

1) Effectiveness of Each Component in the DDC Module: The proposed DDC module combines the use of cosine similarity and Euclidian distance to exploit semantic correlations between query and support images. We conducted separate 1-shot experiments to validate the effects of these two parts, and the experimental results are shown in Table V. It is clear that even using either cosine correlation or Euclidian correlation can lead to good results. Specifically, the model with cosine correlation achieved 34.43% mean mIoU and 56.94% mean FB-IoU, while the model with Euclidean correlation achieved 35.00% mean mIoU and 56.76% mean FB-IoU. Notably, by integrating these two operations, the performance of the model has been significantly improved in all three folds. This indicates that the combination of the two techniques can effectively improve the ability of the model to explore the semantic correspondence between the query and support images.

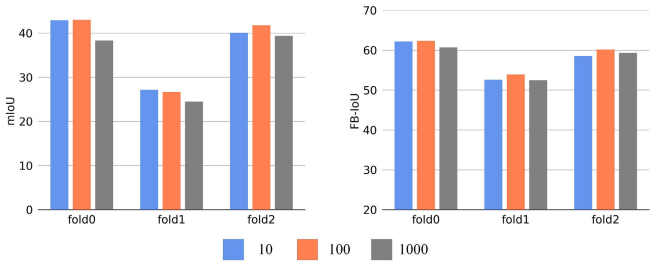


Fig. 5. Ablation study on different K values.

TABLE VI
PERFORMANCE OF THE MODELS TRAINED WITH DIFFERENT LOSS FUNCTIONS

Loss Function	Fold0	Fold1	Fold2	Mean	FB-IoU
Cross-entropy loss	33.70	23.58	34.70	30.66	56.32
Scale-aware focal loss	41.28	26.91	39.44	35.88	57.27
Scale-aware cross-entropy loss	**42.94**	**27.17**	**40.06**	**36.72**	**57.78**

2) Impact of Top K Nearest Matches: The parameter K, which determines the number of top K nearest matches used to generate the Euclidean correlation maps, has the potential to impact the performance of the model. To study this effect, we experimented with different K values (10, 100, and 1000). Fig. 5 shows that the mean mIoU and FB-IoU obtained by the model varied across each fold as K increased. Specifically, when K was set to 100, the overall performance of the model improved significantly, achieving the highest accuracy. However, when we increased K to 1000, the mean mIoU and FB-IoU obtained by the model in the three folds declined severely. This performance degradation from $K = 100$ to 1000 suggests that too much matching information can confuse the classification of objects. In this dataset, numerous objects are smaller than 32×32 pixels, causing the model to take some background features as the guidance of segmentation. As $K = 10$ and $K = 100$ yield similar model performance, we set K to ten uniformly in our experiments to improve the model's inference efficiency.

3) Analysis on Different Loss Functions: To demonstrate the effectiveness of the proposed SACELoss, we conducted experiments comparing models trained with different loss functions. Specifically, we selected the CELoss and SAFLoss [29] for comparison, and the results of these experiments are reported in Table VI. Notably, the model trained using the SACELoss achieved mean mIoU and FB-IoU scores that were 6.06% and 1.46% higher, respectively, than those obtained using the CELoss. Furthermore, the superior performance of the model trained using the proposed loss function is evident across all three folds of the iSAID-5^i dataset, outperforming the model trained using the SAFLoss. To demonstrate the superior performance of our proposed SACELoss function compared to other loss functions, we have conducted a visual comparison of several segmentation results for scenarios involving small targets. The comparison results are depicted in Fig. 6. It is evident that the commonly used CELoss function failed to enable the model to recognize these small targets. For instance, in scenarios where there are numerous small targets in the image, the model tends to miss some of them. However, our proposed loss function assigns a higher

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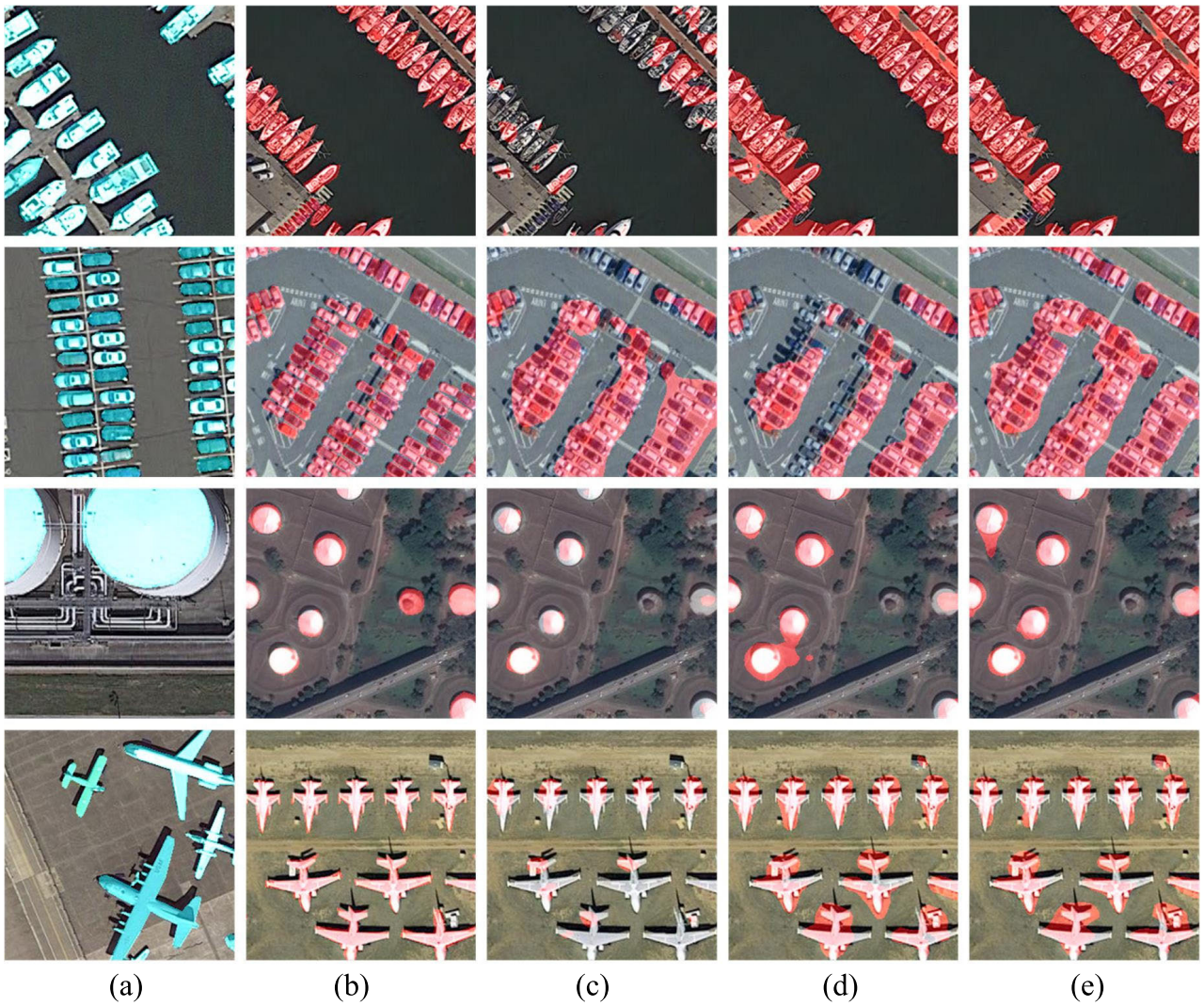


Fig. 6. Qualitative comparison of the segmentation effects of models trained using different loss functions on images that contain small objects. From left to right: (a) support images and their ground truth, (b) query images and their ground truth. Predictions of the model trained with (c) CELoss, (d) SAFLoss, and (e) SACELoss.

weight value to small targets, enabling the model to accurately extract all the targets in the image. Notably, the model trained using our loss function yields more precise object boundaries than the SAFLoss. This is due to the fact that the SAFLoss assigns a higher weight to the entire image, whereas our loss function assigns a higher weight to each individual object. As a result, our method ensures that every tiny object is correctly identified.

4) Analysis on Different Few-Shot Segmentation Strategies: As mentioned in Section III-F, we employed two different few-shot segmentation strategies, where multiple support images were used. We compared the training-free voting method utilized in HSNet [35] with the joint training method that we developed. The results of the 5-shot experiments with VGG16 and ResNet50 are presented in Table VII. Overall, both the two methods can boost the segmentation of the query image. However, our approach achieves higher accuracy with both backbone networks, as it leverages the semantic correspondences among the query image and multiple supporting images. One clear disadvantage of our method is that it requires additional models to be trained for various numbers of support images. On the other hand, the voting method, albeit less accurate, can be employed for any number of support images without the need for additional training.

TABLE VII
COMPARISONS ON DIFFERENT FEW-SHOT SEGMENTATION STRATEGIES

Backbone	Strategy	1-shot		5-shot	
		mIoU	FB-IoU	mIoU	FB-IoU
VGG6	Voting	33.63	55.89	38.94	59.09
	Training			**43.12**	**61.72**
ResNet50	Voting	36.72	57.78	43.06	61.45
	Training			**46.50**	**63.59**

V. CONCLUSION

In this article, we introduce a novel PCFNet designed for the few-shot segmentation of remote-sensing images. Our

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approach includes a DDC module that effectively leverages limited support samples to provide helpful guidance. Additionally, we address the challenge of identifying tiny objects in remote-sensing images by incorporating a SACELoss, which ensures the segmentation accuracy of tiny objects. We conducted comprehensive experiments on the iSAID-5^i and DLRSD-5^i datasets, which demonstrate that our model achieves state-of-the-art performance without compromising efficiency. We believe that our proposed PCFNet can serve as a reliable baseline method for few-shot aerial image semantic segmentation and can also be extended to facilitate multi-scale pixel-wise correspondence in other domains with limited supervision.

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